

Questions with Answer Keys

MathonGo

Q1 - 2024 (27 Jan Shift 1)

Two bodies of mass 4 g and 25 g are moving with equal kinetic energies. The ratio of magnitude of their linear momentum is :

- (1) 3 : 5
- (2) 5 : 4
- (3) 2 : 5
- (4) 4 : 5

Q2 - 2024 (29 Jan Shift 1)

The potential energy function (in J) of a particle in a region of space is given as $U = (2x^2 + 3y^3 + 2z)$. Here x , y and z are in meter. The magnitude of x - component of force (in N) acting on the particle at point

$P(1, 2, 3)m$ is :

- (1) 2
- (2) 6
- (3) 4
- (4) 8

Q3 - 2024 (29 Jan Shift 1)

A block of mass 100 kg slides over a distance of 10 m on a horizontal surface. If the co-efficient of friction between the surfaces is 0.4 , then the work done against friction (in J) is :

- (1) 4200
- (2) 3900
- (3) 4000
- (4) 4500

Q4 - 2024 (29 Jan Shift 1)

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The de-Broglie wavelength of an electron is the same as that of a photon. If velocity of electron is 25% of the velocity of light, then the ratio of K.E. of electron and K.E. of photon will be:

- (1) $\frac{1}{1}$
- (2) $\frac{1}{8}$
- (3) $\frac{8}{1}$
- (4) $\frac{1}{4}$

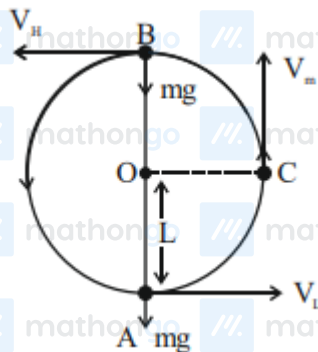
Q5 - 2024 (29 Jan Shift 2)

The bob of a pendulum was released from a horizontal position. The length of the pendulum is 10 m. If it dissipates 10% of its initial energy against air resistance, the speed with which the bob arrives at the lowest point is : [Use, $g : 10 \text{ ms}^{-2}$]

- (1) $6\sqrt{5} \text{ ms}^{-1}$
- (2) $5\sqrt{6} \text{ ms}^{-1}$
- (3) $5\sqrt{5} \text{ ms}^{-1}$
- (4) $2\sqrt{5} \text{ ms}^{-1}$

Q6 - 2024 (29 Jan Shift 2)

A bob of mass ' m ' is suspended by a light string of length ' L '. It is imparted a minimum horizontal velocity at the lowest point A such that it just completes half circle reaching the top most position B. The ratio of kinetic energies $\frac{(K.E.)_A}{(K.E.)_B}$ is :



- (1) 3 : 2

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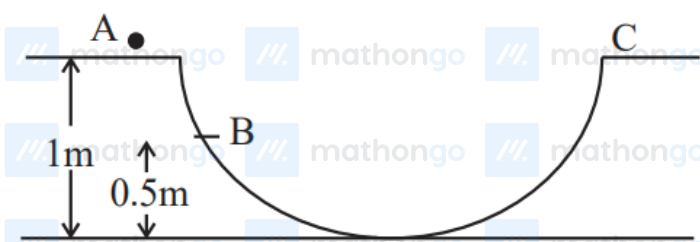
(2) 5 : 1

(3) 2 : 5

(4) 1 : 5

Q7 - 2024 (30 Jan Shift 1)

A particle is placed at the point A of a frictionless track ABC as shown in figure. It is gently pushed toward right. The speed of the particle when it reaches the point B is : (Take $g = 10 \text{ m/s}^2$).



(1) 20 m/s

(2) $\sqrt{10}$ m/s(3) $2\sqrt{10}$ m/s

(4) 10 m/s

Q8 - 2024 (31 Jan Shift 1)

An artillery piece of mass M_1 fires a shell of mass M_2 horizontally. Instantaneously after the firing, the ratio of kinetic energy of the artillery and that of the shell is :

(1) $M_1 / (M_1 + M_2)$ (2) $\frac{M_2}{M_1}$ (3) $M_2 / (M_1 + M_2)$ (4) $\frac{M_1}{M_2}$

Q9 - 2024 (31 Jan Shift 2)

A body of mass 2 kg begins to move under the action of a time dependent force given by

$\vec{F} = (6t\hat{i} + 6t^2\hat{j}) \text{ N}$. The power developed by the force at the time t is given by:

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Questions with Answer Keys

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(1) $(6t^4 + 9t^5) W$	MathonGo	MathonGo	MathonGo	MathonGo	MathonGo
(2) $(3t^3 + 6t^5) W$	MathonGo	MathonGo	MathonGo	MathonGo	MathonGo
(3) $(9t^5 + 6t^3) W$	MathonGo	MathonGo	MathonGo	MathonGo	MathonGo
(4) $(9t^3 + 6t^5) W$	MathonGo	MathonGo	MathonGo	MathonGo	MathonGo
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Questions with Answer Keys

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Q1 (3) mathongo /// math Q**2** (3) math Q**3** (3) math Q**4** (2) math mathongo

Q5 (1) math Q**6** (2) math Q**7** (2) math Q**8** (2) math mathongo

Q9 (4) math math math math math math math math math math math math

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Solutions

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Q1

$$\frac{P_1^2}{2m_1} = \frac{P_2^2}{2m_2}$$

$$\frac{P_1}{P_2} = \sqrt{\frac{m_1}{m_2}} = \frac{2}{5}$$

Q2

Given $U = 2x^2 + 3y^3 + 2z$

$$F_x = -\frac{\partial U}{\partial x} = -4x$$

At $x = 1$ magnitude of F_x is $4N$

Q3

Given $m = 100 \text{ kg}$

$s = 10 \text{ m}$

$\mu = 0.4$

As $f = \mu mg = 0.4 \times 100 \times 10 = 400 \text{ N}$

Now $W = f \cdot s = 400 \times 10 = 4000 \text{ J}$

Q4

For photon

$$E_p = \frac{hc}{\lambda_p} \Rightarrow \lambda_p = \frac{hc}{E_p}$$

For electron

$$\lambda_e = \frac{h}{m_e v_e} = \frac{h v_e}{2 K_e}$$

Given $v_e = 0.25c$

$$\lambda_e = \frac{h \times 0.25c}{2 K_e} = \frac{hc}{8 K_e}$$

Also $\lambda_p = \lambda_e$

$$\frac{hc}{E_p} = \frac{hc}{8 K_e}$$

$$\frac{K_e}{E_p} = \frac{1}{8}$$

Q5

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Solutions

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$$\ell = 10 \text{ m}$$

$$\text{Initial energy} = mg\ell$$

$$\text{So, } \frac{9}{10}mg\ell = \frac{1}{2}mv^2$$

$$\Rightarrow \frac{9}{10} \times 10 \times 10 = \frac{1}{2}v^2$$

$$v^2 = 180$$

$$v = \sqrt{180} = 6\sqrt{5} \text{ m/s}$$

Q6

Apply energy conservation between A & B

$$\frac{1}{2}mV_L^2 = \frac{1}{2}mV_H^2 + mg(2L)$$

$$\therefore V_L = \sqrt{5gL}$$

$$\text{So, } V_H = \sqrt{gL}$$

$$\frac{(K.E)_A}{(K.E)_B} = \frac{\frac{1}{2}m(\sqrt{5gL})^2}{\frac{1}{2}m(\sqrt{gL})^2} = \frac{5}{1}$$

Q7

By COME

$$KE_A + U_A = KE_B + U_B$$

$$0 + mg(1) = \frac{1}{2}mv^2 + mg \times 0.5$$

$$v = \sqrt{g} = \sqrt{10} \text{ m/s}$$

Q8

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Solutions

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$$|\vec{p}_1| = |\vec{p}_2|$$

$$KE = \frac{p^2}{2M}; p \text{ same}$$

$$KE \propto \frac{1}{m}$$

$$\frac{KE_1}{KE_2} = \frac{p^2/2M_1}{p^2/2M_2} = \frac{M_2}{M_1}$$

Q9

$$\vec{F} = (6t\hat{i} + 6t^2\hat{j}) \text{ N}$$

$$\vec{F} = ma = (6t\hat{i} + 6t^2\hat{j})$$

$$\vec{a} = \frac{\vec{F}}{m} = (3t\hat{i} + 3t^2\hat{j})$$

$$\vec{v} = \int_0^t \vec{a} dt = \frac{3t^2}{2}\hat{i} + t^3\hat{j}$$

$$P = \vec{F} \cdot \vec{v} = (9t^3 + 6t^5) \text{ W}$$

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